

Daewon Chae

✉ daewon@umich.edu

🌐 <https://daewon88.github.io/>

Research Interest

My current research interests focus on improving the efficiency of AI models. In particular, I am interested in reducing model complexity through structured matrix methods such as low-rank approximations, as well as related topics including quantization. Previously, I worked on reinforcement learning for generative model post-training.

Education

University of Michigan, Ann Arbor

Sep. 2025 - Present

Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Hun-Seok Kim)

Korea University

Mar. 2023 - Aug. 2025

M.S. in Computer Science and Engineering (Advisor: Prof. Jinkyu Kim)

Korea University

Mar. 2017 - Aug. 2022

B.S. in Electrical Engineering

Publications

* : equal contributions

- **DyRA: Dynamic Residual Approximation for Efficient Matrix Multiplication in DNNs**

Daewon Chae, Hyunwon Chung, Changwoo Lee, Hun-Seok Kim

Neural Information Processing Systems (NeurIPS), 2026

- **Streaming Dense Voxel Representations for 3D Occupancy Prediction**

Seokha Moon, Janghyun Baek, Yujin Jeong, Daewon Chae, Giseop Kim, Jungbeom Lee, Jinkyu Kim, Sunwook Choi

European Conference on Computer Vision (ECCV), 2026

- **Grounding Robot Action Coordinate System in RGB Observations for Robust Visuomotor Manipulation**

Jiyun Jang, Yujin Sung, Woosung Joung, Daewon Chae, Sangwon Lee, Sohwi Kim, Jinkyu Kim, Jungbeom Lee

Robotics: Science and Systems (RSS), 2026

- **ENTP: Encoder-only Next Token Prediction**

Ethan Ewer*, Daewon Chae*, Thomas Zeng*, Jinkyu Kim, Kangwook Lee

Transactions on Machine Learning Research (TMLR), 2025

NeurIPS Workshop on What Can('t) Transformers Do? (Spotlight Presentation), 2025

- **Hard-Negative Prototype-Based Regularization for Few-Shot Class-Incremental Learning**

Seongbeom Park, Hyunju Yun, Daewon Chae, Sungyoon Kim, Suhong Moon, Minwoo Kang, Seunghyun Park, Jinkyu Kim

Transactions on Machine Learning Research (TMLR), 2025

- **VersaPRM: Multi-Domain Process Reward Model via Synthetic Reasoning Data**

Thomas Zeng, Shuibai Zhang, Shutong Wu, Christian Classen, Daewon Chae, Ethan Ewer, Minjae Lee, Heeju Kim, Wonjun Kang, Jackson Kunde, Ying Fan, Jungtaek Kim, Hyung Il Koo, Kannan Ramchandran, Dimitris Papailiopoulos, Kangwook Lee

International Conference on Machine Learning (ICML) (Oral Presentation), 2025

- **DiffExp: Efficient Exploration in Reward Fine-tuning for Text-to-Image Diffusion Models**

Daewon Chae*, June Suk Choi*, Jinkyu Kim, Kimin Lee

AAAI Conference on Artificial Intelligence (AAAI) (Oral Presentation), 2025

- **A Training-Free Approach for Handling Loosely Aligned Visual Conditions in ControlNet**

Woosung Joung*, Daewon Chae*, Jinkyu Kim

British Machine Vision Conference (BMVC), 2025

- **Scene Graph-Guided Proactive Replanning for Failure-Resilient Embodied Agent**

Che Rin Yu, Daewon Chae, Dabin Seo, Sangwon Lee, Hyeongwoo Im, Jinkyu Kim

Autonomous Agents and Multiagent Systems (AAMAS), 2026

RSS Workshop on Semantic Reasoning and Goal Understanding in Robotics (RSSW), 2025

- **InstructBooth: Instruction-following Personalized Text-to-Image Generation**

Daewon Chae, Nokyoung Park, Jinkyu Kim, Kimin Lee

ICML Workshop on Foundation Models in the Wild (ICMLW), 2024

- **Clustering-based Image-Text Graph Matching for Domain Generalization**

Nokyoung Park, Daewon Chae, Jeongyong Shim, Sangpil Kim, Eun-Sol Kim, Jinkyu Kim

International Conference on Pattern Recognition (ICPR), 2024

- **Text-driven Prototype Learning for Few-Shot Class-Incremental Learning**

Seongbeom Park*, Haeji Jung*, Daewon Chae, Hyunju Yun, Sungyoon Kim, Suhong Moon, Seunghyun Park, Jinkyu Kim

International Conference on Pattern Recognition (ICPR), 2024

- **Re-ID Technology for V2I based Cooperative Driving Protocol**

Junhyek Jang, Kitaeg Lim, Sanghun Yoon, Daewon Chae, Soohyun Jang

International Conference on Ubiquitous and Future Networks (ICUFN), 2023

Experience

University of Wisconsin-Madison, Lee Lab

Jul. 2024 - Sep. 2024

Visiting Researcher (Advisor: Prof. Kangwook Lee)

Korea Electronics Technology Institute

Oct. 2022 - Feb. 2023

Assistant Researcher (Mentor: Soohyun Jang)

NAVER Clova, High-quality DNN Text-to-Speech (HDTS) Team,

Sep. 2021 - Feb. 2022

Research Intern (Mentor: Eunwoo Song)

Seoul National University, Music and Audio Research Group

Dec. 2020 - Feb. 2021

Research Intern (Mentor: Juheon Lee)

Honors and Awards

Scholarship

Hyundai On-dream Future Industrial Talent Scholarship

2023 - 2024

Hyundai Motor Chung Mong-Koo Foundation

- Full tuition and living expenses support for graduate studies

Cheonman Scholarship

2017 - 2022

Cheonman Scholarship Foundation

- Full tuition and living expenses support for undergraduate studies

Awards

Image Sound matching AI Competition, 2nd place (\$1500 as awards)

2022

National Information Society Agency (NIA)

Korean STT AI Competition, 5th place (\$3000 as awards)

2022

National Information Society Agency (NIA) and Hyundai Motor

Multi-Camera Multi-Object Tracking Challenge, 3rd place (\$500 as awards)
Korean Conference on Computer Vision (KCCV) Workshop organized by 42dot

2022

Teaching & Leadership Experience

Teaching Assistant: Data Science (COSE471), Korea University

Spring 2023

Club President: EE programming club (One and Zero), Korea University

2020 - 2021

Academic Services

Reviewer: CoRL2025, ICML2025, NeurIPS 2025, AISTATS 2025,